example just given, the person would actually have brown eyes: the instructions for making blue eyes would be ignored in the building of the body, though this does not stop them being passed on to future generations. A gene that is ignored in this way is called *recessive*. The opposite of a recessive gene is a *dominant* gene. The gene for brown eyes is dominant to the gene for blue eyes. A person has blue eyes only if both copies of the relevant page are unanimous in recommending blue eyes. More usually when two alternative genes are not identical, the result is some kind of compromise—the body is built to an intermediate design or something completely different.

When two genes, like the brown eye and the blue eye gene, are rivals for the same slot on a chromosome, they are called *alleles* of each other. For our purposes, the word allele is synonymous with rival. Imagine the volumes of architects' plans as being loose-leaf binders, whose pages can be detached and interchanged. Every Volume 13 must have a Page 6, but there are several possible Page 6s which could go in the binder between Page 5 and Page 7. One version says 'blue eyes', another possible version says 'brown eyes'; there may be yet other versions in the population at large which spell out other colours like green. Perhaps there are half a dozen alternative alleles sitting in the Page 6 position on the 13th chromosomes scattered around the population as a whole. Any given person only has two Volume 13 chromosomes. Therefore he can have a maximum of two alleles in the Page 6 slot. He may, like a blue-eyed person, have two copies of the same allele, or he may have any two alleles chosen from the half dozen alternatives available in the population at large.

You cannot, of course, literally go and choose your genes from a pool of genes available to the whole population. At any given time all the genes are tied up inside individual survival machines. Our genes are doled out to us at conception, and there is nothing we can do about this. Nevertheless, there is a sense in which, in the long term, the genes of the population in general can be regarded as a *gene pool*. This phrase is in fact a technical term used by geneticists. The gene pool is a worthwhile abstraction because sex mixes genes up, albeit in a carefully organized way. In particular, something like the detaching and interchanging of pages and wads of pages from loose-leaf binders really does go on, as we shall presently see.